

Shreejit Verma Homework 3

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Soln Suppose that X is a positive random variable. We know that $E(1/X) \neq 1/E(X)$, in general. Can you claim that either $E(1/X) \geq 1/E(X)$ or $E(1/X) \leq 1/E(X)$? Explain your response.

Soln No, we can't claim either because if the function is convex

$$f(x) = \frac{1}{x}$$

$$E\left(\frac{1}{x}\right) \geq \frac{1}{E(x)}$$
 and if it is concave then,

$$E\left(\frac{1}{x}\right) \leq \frac{1}{E(x)}$$

23. Consider a circle of radius 1 centered at the origin, and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let X and Y be the coordinates of the point chosen, it follows that the joint density function of X and Y is given by:

$$f_{XY}(x, y) = \begin{cases} 1/\pi & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginals of X and Y . (They are the same due to symmetry.)
 (b) Let D be the distance between the origin to the point selected. Find the probability that D is less or equal to d . In other words, find $F_D(d)$.
Guidance: Note that $D = \sqrt{X^2 + Y^2}$.

(c) Find $E(D)$.

Soln a) $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$

$$= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy$$

$$= \frac{1}{\pi} [y]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}$$

$$= \frac{1}{\pi} (\sqrt{1-x^2} - (-\sqrt{1-x^2}))$$

$$= \frac{1}{\pi} \times 2\sqrt{1-x^2}$$

$$\text{Similarly } f_Y(y) = \frac{1}{\pi} \times 2\sqrt{1-y^2}$$

b) $D = \sqrt{x^2 + y^2}$
 distance from origin $0 \leq d \leq 1$

$$F_D(d) = P(\sqrt{x^2 + y^2} \leq d)$$

$$F_D(d) = P(x^2 + y^2 \leq d^2)$$

$$F_D(d) = \iint_{x^2 + y^2 \leq d^2} \frac{1}{\pi} dx dy$$

$$F_D(d) = \frac{1}{\pi} \iint_{x^2 + y^2 \leq d^2} dx dy$$

$$F_D(d) = \frac{1}{\pi} [\pi d^2]$$

$$F_D(d) = d^2$$

c) To find $E(D)$ we use cumulative distribution function of D and differentiate it to find PDF.

$F_D(d) = d^2$ then derivative of $F_D(d)$ w.r.t d will give us pdf:

$$f_D(d) = 2d \quad \text{for } 0 \leq d \leq 1$$

$$E(D) = \int_{-\infty}^{\infty} d f_D(d) dd$$

$$E(D) = \int_0^1 d \cdot 2d \cdot dd$$

$$E(D) = 2 \int_0^1 d^2 dd$$

$$E(D) = 2 \left[\frac{d^3}{3} \right]_{d=0}^{d=1} = 2 \times \frac{1}{3} = \frac{2}{3} \quad \text{Ans}$$

26. Let $X \sim \text{Exp}(\lambda_1)$ and that $Y \sim \text{Exp}(\lambda_2)$.

If we assume that X and Y are independent, then the joint pdf of X and Y is

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) = \begin{cases} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

- (a) For $t > 0$, prove that $P(Y < tX) = \frac{t \lambda_2}{\lambda_1 + t \lambda_2}$
 (b) For simplicity, assume that $\lambda_1 = \lambda_2 = 1$, and let $T = \frac{Y}{X}$.
 Use part (a) to find the pdf of T , $f_T(t)$.
 (c) Without doing any formal derivations beyond what was done in part (b), explain why $E(T) = \infty$.

30. Let (X, Y) have the following joint pdf:

$$f_{XY}(x, y) = \begin{cases} 60x^2y & x > 0, y > 0, x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal distributions $f_X(x)$ and $f_Y(y)$.
 (b) Use properties of the Beta distribution to find $E(X)$, $E(Y)$, $V(X)$, and $V(Y)$.
 (c) Find $\text{Cov}(X, Y)$ and $\text{Corr}(X, Y)$.
 (d) Find $V(X + Y)$.

Soln a) $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$

$$= \int_0^{1-x} 60x^2y dy$$

$$= \left[\frac{60x^2y^2}{2} \right]_0^{1-x}$$

$$= \frac{60x^2(1-x)^2}{2} = 30x^2(1-x)^2$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

$$= \int_0^{1-y} 60x^2y dx$$

$$= \left[\frac{60x^3}{3} y \right]_0^{1-y}$$

$$= 20(1-y)^2 y$$

b) $E(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x, y) dx dy$ B 33
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$$= \int_0^1 \int_0^{1-y} x \cdot 60x^2y dx dy$$

$$= \int_0^1 \int_0^{1-y} 60x^3y dx dy$$

$$= \int_0^1 \int_0^{1-y} y \cdot 60x^3y dx dy$$

32. Suppose that X_1, X_2, X_3 have the following joint pdf:

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} 6e^{-(x_1 + 2x_2 + 3x_3)} & x_i > 0, i = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

Are X_1, X_2 , and X_3 independent? Justify your answer rigorously.

Soln Yes, As $f_{X_1}(x_1) f_{X_2}(x_2) f_{X_3}(x_3) = f_{X_1, X_2, X_3}(x_1, x_2, x_3)$

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 6e^{-(x_1 + 2x_2 + 3x_3)} dx_1 dx_2 dx_3$$

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 6e^{-(x_1 + 2x_2 + 3x_3)} dx_2 dx_3$$

$$f_{X_2}(x_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 6e^{-(x_1 + 2x_2 + 3x_3)} dx_1 dx_3$$

$$f_{X_3}(x_3) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 6e^{-(x_1 + 2x_2 + 3x_3)} dx_1 dx_2$$